

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0509

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

1. Grey Level Transformation

- (a) Explain the purpose of grey level transformation in image enhancement.
- (b) Apply a negative transformation $s = 255 - r$ to the pixel values:

$$r = [50, 100, 150, 200]$$

Compute the transformed values.

2. Log Transformation

- (a) Explain how log transformation enhances low-intensity pixels.
- (b) Given $s = c \log(1 + r)$, where $c = 10$, compute s for:

$$r = [0, 10, 50, 100]$$

3. Histogram Processing

- (a) Explain the role of histogram equalization in contrast enhancement.
- (b) Given a 3-bit image with histogram:

Intensity Frequency

0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.

4. Spatial Filtering (Smoothing)

- (a) Explain the purpose of spatial filtering for smoothing.
- (b) Apply a 3×3 averaging filter to the following region:

98
10

[10 20 30 20 30 40 30 40 50]

Compute the new center pixel value.

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

to the matrix:

$$[10 \ 10 \ 10, 10 \ 20 \ 10, 10 \ 10 \ 10]$$

Compute the output at the center.

6. Frequency Domain Filtering

(a) Differentiate low-pass and high-pass filtering in frequency domain.

(b) Given frequency components:

$$F = [100, 80, 40, 20]$$

Apply a low-pass filter that retains only the first two components. Write the filtered output.

7. Thresholding

(a) Explain global thresholding in segmentation.

(b) Given pixel values:

$$[200]$$

Apply threshold $T = 100$. Assign values (0 or 1).

8. Edge Detection

(a) Explain the concept of edge detection.

(b) Apply the Sobel operator (horizontal):

$$\begin{bmatrix} -1 & -2 & -1 & 0 & 0 & 0 & 1 & 2 & 1 \end{bmatrix}$$

to:

$$[10 \ 10 \ 10, 20 \ 20 \ 20, 30 \ 30 \ 30]$$

Compute the gradient at the center.

9. Region-Based Segmentation

(a) Explain region growing in segmentation.

(b) Given pixel intensities:

$$[1 \ 0 \ 2]$$

If seed = 50 and threshold difference = 5, identify pixels belonging to the region.

10. Combined Enhancement & Segmentation

(a) Explain why smoothing is applied before segmentation.

(b) Given image values:

[1 0 5]

First apply averaging (assume mean = 48), then threshold at $T = 50$.

Determine final segmented output.

11. Power-Law (Gamma) Transformation

A grayscale image is enhanced using power-law transformation.

(a) Interpret the role of gamma correction in image enhancement.

(b) Justify the use of transformation $s = cr^\gamma$, where $\gamma = 0.5$, $c = 1$.

(c) Apply the transformation step-by-step to pixel values:

$r = [25, 100, 225]$.

(d) Compute the transformed values accurately.

(e) Interpret the results in terms of brightness enhancement.

12. Contrast Stretching

An image has intensity values in the range [50, 150].

(a) Interpret the need for contrast stretching.

(b) Justify linear transformation to range [0, 255].

(c) Apply transformation step-by-step for values:

$r = [50, 100, 150]$.

(d) Compute the output values accurately.

(e) Interpret improvement in contrast.

13. Histogram Equalization (4-bit Image)

A 4-bit image has the following histogram:

Intensity Frequency

0 4

1 6

2 10

3 6

(a) Interpret the need for equalization.

(b) Justify use of CDF.

(c) Compute cumulative distribution.

(d) Compute new intensity levels.

(e) Interpret contrast enhancement.

14. Median Filtering

A noisy image contains salt-and-pepper noise.

(a) Interpret noise characteristics.

(b) Justify median filter usage.

(c) Apply filter to:

10 200 10, 200 50 200, 10 200 10

(d) Compute center value.

(e) Interpret noise removal effectiveness.

15. Gaussian Smoothing

An image is smoothed using Gaussian filter.

(a) Interpret smoothing objective.

(b) Justify Gaussian filter over averaging.

(c) Apply filter:

[1 2 1, 2 4 2, 1 2 1]

to matrix:

[10 20 10, 20 40 20, 10 20 10]

(d) Compute center pixel.

(e) Interpret smoothing effect.

Code of Conduct:

I N. Akshaya (192425129) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N. Akshaya

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Assessment Tool-1(CO2)

N. Akshaya

192425129

ITA0509

Computer Vision.

1) Grey Level Transformation

a) Explain the purpose of grey level transformation in image enhancement.

Grey Level transformation is used to improve the appearance of an image by changing the intensity values of pixels. It enhances image contrast, highlights important features, and makes objects easier to identify. The negative transformation is particularly useful for displaying white or gray details on a dark background.

b) Apply the negative transformation $s = 255 - r$ for $r = [50, 100, 150, 200]$.

Formula:

$$s = 255 - r$$

Original pixel (r)	Calculation	Transformed Pixel (s)
50	$255 - 50$	205
100	$255 - 100$	155
150	$255 - 150$	105
200	$255 - 200$	55

Answer: [205, 155, 105, 55]

Interpretation: The transformation reverse the brightness levels. Dark pixels becomes bright and bright pixels become dark, improving the visibility of hidden details.

2) Log Transformation

a) Explain how log Transformation enhances low-intensity pixels.

Log transformation increases the brightness of low-intensity pixels more than high-intensity pixels. As a result, dark regions are compressed, producing better overall image visibility.

b) Compute $s = C \log(1+r)$, where $C=10$.

Formula:

$$s = 10 \log(1+r)$$

using natural logarithm (ln):

r	Calculation	s
0	$10 \times \ln(1)$	0.00
10	$10 \times \ln(11)$	23.98
50	$10 \times \ln(51)$	39.32
100	$10 \times \ln(101)$	46.15

Answer:

[0.00, 23.98, 39.32, 46.15]

Interpretation: Low-Intensity pixels are enhanced more than high-intensity pixels, making dark portions of the image easier to observe.

Histogram processing

a) Explain the role of histogram equalization in contrast enhancement.

Histogram equalization improves image contrast by redistributing pixels intensity values across the available range. This makes details in both dark and bright regions more visible.

b) Compute the cumulative distribution and equalized intensity values.

Histogram:

Intensity	Frequency
0	2
1	2
2	4
3	8

Total pixels:

$$N = 2 + 2 + 4 + 8 = 16$$

Cumulative Distribution Function (CDF):

Intensity	CDF
0	$2/16 = 0.125$
1	$4/16 = 0.25$
2	$8/16 = 0.50$
3	$16/16 = 1.00$

For a 3-bit image,

$$L = 8$$

$$\text{Equalized intensity: } S = (L-1) \times \text{CDF} = 7 \times \text{CDF}$$

Intensity	Equalized Value
0	1
1	2
2	4
3	7

Answer: [1, 2, 4, 7]

Interpretation: The intensity values are spread over a wider range, increasing image contrast and improving visibility of image details.

4) Spatial Filtering (Smoothing)

a) Explain the purpose of spatial filtering for Smoothing.

Spatial Smoothing is used to reduce random noise and small intensity variations in an image. It produces a smoother image and improves image quality before further processing as gen Segmentation.

b) Apply a 3x3 averaging filter.

Given matrix:

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{bmatrix}$$

Sum of all pixels:

$$10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$$

$$\text{Average: } 270 / 9 = 30$$

Answer: 30

Interpretation: The new center pixel value is 30, which smooths the image by reducing small intensity variations and noise.

5) Spatial Filtering (Sharpening)

a) Explain how Sharpening enhances image details.

Sharpening improves image quality by emphasizing edges and fine details. It increases the contrast between neighbouring pixels, making object boundaries more distinct.

b) Apply the Laplacian mask.

Mask:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Image:

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Calculation:

$$(0 \times 10) + (-1 \times 10) + (0 \times 10) + (-1 \times 10) + (4 \times 20) + (-1 \times 10) + (0 \times 10) + (-1 \times 10) + (0 \times 10)$$

$$= 0 - 10 + 0 - 10 + 80 - 10 + 0 - 10 + 0$$

$$= 40$$

Answer: 40

Interpretation: The out value 40 indicates that the center pixel is enhanced, making edges and image details clearer. This improves image sharpness and boundary detection.

6) Frequency Domain Filtering

a) Differentiate low-pass and high-pass filtering in the frequency domain.

Low-pass Filter	High-pass Filter
Allows low-Frequency Components to pass.	Allows high-Frequency Components to pass.
Removes noise and smooths the image.	Enhances edges & fine details
Produces a blurred image	produces a sharper image.

b) Apply a low-pass filter to the frequency components:

Given:

$F = [100, 80, 40, 20]$

A low-pass filter retains only the first two frequency components.

Filtered output:

$[100, 80, 0, 0]$

Interpretation:

The high-frequency components are removed, reducing noise and producing a smoother image.

7) Thresholding

a) Explain global thresholding in segmentation.

Global thresholding is an image segmentation technique that separates the foreground from the background using a single threshold value. Pixels with intensity greater than or equal to the threshold are assigned 1, while pixels below the threshold are assigned 0.

b) Apply threshold $T=100$.

Given pixel value:

$[200]$

Rule:

* pixel $\geq 100 \rightarrow 1$

* pixel $< 100 \rightarrow 0$

Since

$200 > 100$

output: [1]

Interpretation: The pixel is classified as the foreground because its intensity is greater than the threshold.

8) Edge Detection

a) Explain the concept of edge detection.

Edge detection identifies sudden changes in pixel intensity within an image. It is used to detect object boundaries, shapes, and important image features for segmentation and recognition.

b) Apply the Sobel horizontal operator.

Sobel mask:

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Image:

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

Calculation:

$$\text{Top row: } (-1 \times 10) + (-2 \times 10) + (-1 \times 10) = -10 - 20 - 10 = -40$$

$$\text{Middle row: } 0$$

Bottom row:

$$(1 \times 30) + (2 \times 30) + (1 \times 30) = 30 + 60 + 30 = 120$$

$$\text{Gradient: } 120 - 40 = 80$$

Answer: 90

Interpretation: A gradient value of 80 indicates a strong horizontal edge at the centre of the image.

Q Region-Based Segmentation

a) Explain region growing in segmentation.

Region growing is a segmentation method that starts from a selected seed pixel and groups neighbouring pixels whose intensity values are similar to the seed within a specified threshold.

b) Identify pixels belonging to the region.

Given pixel intensities:

[48, 50, 52]

Seed value: 50

Threshold difference: 5

Calculate the intensity difference:

Pixel	Difference from 50
48	2
50	0
52	2

All difference are less than or equal to 5.

∴ all pixels belong to the region.

Answer:

[48, 50, 52]

Interpretation:

All three pixels are similar to the seed intensity and are grouped into the same segmented region.

Combined Enhancement & Segmentation.

a) Explain why Smoothing is applied before Segmentation.

Smoothing is applied before Segmentation to reduce image noise and unwanted intensity variations. This improves Segmentation accuracy by preventing false boundaries and producing more reliable regions.

b) Apply averaging followed by thresholding.

Given:

Average intensity after Smoothing: 48

Threshold:

$$T = 50$$

Rule:

$$* \text{ pixel } \geq 50 \rightarrow 1$$

$$* \text{ pixel } < 50 \rightarrow 0$$

Since:

$$48 < 50$$

Segmented output: 0

Interpretation:

The smoothed pixel value is below the threshold, so it is classified as the background. Noise reduction before thresholding results in more accurate segmentation.

11) power-law (Gamma) Transformation

a) Interpret the role of gamma correction in image enhancement.

Gamma correction adjusts the brightness of an image by modifying pixel intensity values using a power-law function.

It improves image visibility by enhancing on the gamma value
b) Justify the use of the transformation where $s = cr^{\gamma}$, where
 $C=1$ and $\gamma=0.5$

Since $\gamma=0.5 < 1$, the transformation increases the brightness of low-intensity pixels. This is useful for enhancing dark images while preserving image details.

c) Apply the transformation step-by-step to pixel values $r = [25, 100, 225]$.

Given:

$$C = 1$$

$$\gamma = 0.5$$

$$\text{Formula: } s = cr^{\gamma} = 1 \times r^{0.5} = \sqrt{r}$$

$$\text{For } r = 25$$

$$s = \sqrt{25} = 5$$

$$\text{For } r = 100$$

$$s = \sqrt{100} = 10$$

$$\text{For } r = 225:$$

$$s = \sqrt{225} = 15$$

d) Compute the transformed values accurately.

original pixel(r)	Transformed pixel(s)
25	5
100	10
225	15

Answer: 5, 10, 15

e) Interpretation

The transformed values increase the brightness of darker pixels, improving image visibility and revealing details in dark regions.

value.

Contrast Stretching

a) Interpret the need for contrast stretching.

Contrast stretching improves image quality by expanding a narrow range of intensity values to the full grayscale range. This makes image features more distinguishable.

b) Justify linear transformation to range $[0, 255]$.

Linear contrast stretching maps the minimum intensity to 0 and the maximum intensity to 255, thereby utilizing the complete grayscale range and improving contrast.

c) Apply the transformation step-by-step for values

$$r = [50, 100, 150].$$

Given:

* Minimum intensity, $r_{\min} = 50$

* maximum intensity, $r_{\max} = 150$

Formula:

$$s = \frac{(r - r_{\min})}{(r_{\max} - r_{\min})} \times 255$$

For $r = 50$:

$$s = \frac{50 - 50}{150 - 50} \times 255 = \frac{0}{100} \times 255 = 0$$

For $r = 100$:

$$s = \frac{100 - 50}{150 - 50} \times 255 = \frac{50}{100} \times 255 = 127.5 \approx 128$$

For $r = 150$:

$$s = \frac{150 - 50}{150 - 50} \times 255 = \frac{100}{100} \times 255 = 255$$

d) Compute the output values accurately.

Original intensity (r)	Transformed Intensity (s)
50	0
100	128
150	255

Answer: [0, 128, 255]

e) Interpretation

The intensity values now occupy the complete grayscale range, resulting in improved image contrast and clearer visual details.

13) Histogram Equalization

Histogram:

Intensity	Frequency
0	4
1	6
2	10
3	6

a) Interpret the need for histogram equalization.

Histogram Equalization improves image contrast by distributing intensity values more evenly across the available grayscale range.

b) Justify the use of CDF

The Cumulative Distribution Function (CDF) maps old intensity values based on the cumulative probability of occurrence, ensuring better contrast.

c) Compute Cumulative distribution.

Total pixels:

$$N = 4 + 6 + 10 + 6 = 26$$

Intensity	Frequency	CDF
0	4	$4/26 = 0.154$
1	6	$10/26 = 0.385$
2	10	$20/26 = 0.769$
3	6	$26/26 = 1.000$

d) Compute new intensity levels.

For a 4-bit image

$$L = 16$$

Maximum intensity: $L - 1 = 15$

New intensity
 $s = 15 \times \text{CDF}$

Intensity	New Intensity
0	2
1	6
2	12
3	15

Answer:

$[2, 6, 12, 15]$

e) Interpretation:

The equalized image has improved contrast because the intensity values are distributed over a wider grayscale range.

14) Median Filtering

a) Interpret noise characteristics

Salt-and-pepper noise appears as randomly occurring black and white pixels in an image. It reduces image quality & affects edge detection.

b) Justify median filter usage.

A median filter removes salt-and-pepper noise while preserving image edges better than an averaging filter.

c) Apply the median filter.

Given matrix:

$$\begin{bmatrix} 10 & 200 & 10 \\ 200 & 50 & 200 \\ 10 & 200 & 10 \end{bmatrix}$$

Arrange values in ascending order:

10, 10, 10, 10, 50, 200, 200, 200, 200

Median (5th value): 50

d) Center value: 50

e) Interpretation:

The median filter removes impulse noise while maintaining important image edges, producing a cleaner image.

Gaussian Smoothing

a) Interpret the Smoothing objective.

Gaussian Smoothing reduces image noise and small intensity variations while preserving important image structures better than simple averaging.

b) Justify Gaussian filter over averaging.

The Gaussian filter assigns higher weights to the centre pixels and lower weights to neighbouring pixels. This provides smoother results with less blurring than an averaging filter.

c) Apply the Gaussian filter.

Kernel:

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Image:

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

Weighted Sum:

$$\begin{aligned} & (1 \times 10) + (2 \times 20) + (1 \times 10) + (2 \times 20) + (4 \times 40) + (2 \times 20) + (1 \times 10) + (2 \times 20) \\ & + (1 \times 10) \end{aligned}$$

$$\Rightarrow 10 + 40 + 40 + 40 + 160 + 40 + 10 + 40 + 10$$

$$\Rightarrow 360$$

= Sum of kernel weights:

$$1 + 2 + 1 + 2 + 4 + 2 + 1 + 2 + 1 = 16$$

$$\text{Center pixel: } \frac{360}{16} = 22.5$$

d) output: 22.5

e) Interpretation

The Gaussian filter smooths the image by reducing noise while preserving important image details & edges more effectively than a simple averaging filter.

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$(0.22) + (0.11) + (0.22) + (0.11) + (0.22) + (0.11) + (0.22) + (0.11) + (0.22)$$

$$2.22$$

$$2.22$$

$$2.22$$

$$\text{Filter kernel: } \frac{2.22}{16}$$

$$\text{Output: } 2.22$$